ZCAS UNIVERSITY

CIT1742 PROFESSIONAL ETHICS IN COMPUTING

MID-SEMESTER EXAMINATION

FRIDAY, 27TH OCTOBER, 2023

08:30 HRS-11:30HRS

TIME ALLOWED: WRITING – THREE (3) HOURS

READING – FIVE (5) MINUTES

INSTRUCTIONS TO CANDIDATES:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

Samantha is a software developer working for a technology company that specializes in creating medical information systems. Recently, she was tasked to work on a system for a Lusaka based Medical Centre. The system allows patients to schedule appointments with tele-doctors and make online payments for medical services. Samantha has discovered a critical security vulnerability in the company's medical information system. This vulnerability could potentially be exploited by cybercriminals, leading to data breaches and harm to the patients and healthcare providers who rely on the system. However, the company is on the verge of launching a new product, and fixing the vulnerability would delay the release and incur significant costs. Samantha's manager is pressuring her to keep the issue quiet and proceed with the launch, arguing that the impact of the vulnerability is uncertain.

a) Discuss the possible consequences of Samantha following her boss's advice. (10 marks)

b) Evaluate the scenario from a Kantian perspective. (10 marks)

c) Evaluate the scenario from an act utilitarian perspective. (10 marks)

d) Evaluate the scenario from social contract perspective. (10 marks)

(Total: 40 marks)

SECTION B: Attempt any THREE questions in this section

Question 2

a) Define ethics, and explain why it is important for organizations to uphold ethical principles? (5 marks)

b) Compare and contrast the decision-making approaches, decision-makers, consequential focus, and ethical principles or bases of the following pairs of ethical theories:

i. Social Contract Theory and cultural relativism (5 marks)

ii. Ethical Egoism and Subjective Relativism (5 marks)

iii. Kantianism and Divine command theory (5 marks)

(20 marks)

Question 3

a) Complete the table below (12 marks)

Type of Intellectual Property	What It Protects	Duration of Protection	Who/how it Protects
Copyright			
Trademark			
Patent			
Trade Secret			

b) Imagine you are a content creator who has recently uploaded a video to your YouTube channel. In the video, you included a short clip from a popular movie to illustrate a point you were making about film history. You didn't seek permission from the movie studio, but you believe that your use of the clip falls under the fair use doctrine. A few weeks later, you receive a copyright takedown notice from the movie studio, demanding that your video be removed.

Explain how you would you apply the principles of fair use to evaluate whether your use of the movie clip is justifiable? (08 marks)

(20 marks)

Question 4

The Facebook-Cambridge Analytica scandal was a major data privacy controversy that emerged in 2018. It involved the unauthorized and improper harvesting of personal data from approximately 87 million Facebook users by the British political consulting firm Cambridge Analytica. The data was collected through a Facebook app created by a Cambridge University researcher, Dr. Aleksandr Kogan, in 2014. This app not only collected data from its users but also gathered data from their Facebook friends, often without their knowledge or consent.

Cambridge Analytica used the harvested data to create psychological profiles of users, which were then used for targeted political advertising and messaging. It led to public outrage, regulatory investigations, and legal actions.

a) Discuss potential issues or challenges that could emerge as a consequence of the scandal. (6 marks)

b) Identify what data protection principles were breached or compromised in this scenario. (6 marks)

c) When developing apps for public use, software developers often adhere to a code of conduct. What are some of the responsibilities and obligations that developers must consider when releasing software to the public? (8 marks)

(20 marks)

Question 5

Professionalism encompasses a number of different attributes, and, together, these attributes identify and define a professional.

a) Explain in your own words what professionalism is and discuss five (5) attributes of a professional. (6 marks)

b) Discuss what green computing and demonstrate with examples how organisations can practice Green IT practices. (8 marks)

c) Differentiate Compliance-Based and Value-Based Code of Ethics. (6 marks)

(20 marks)

END OF EXAMINATION